Roll No. | | | | | | | | | |

Total No. of Questions : 13

Total No. of Pages : 02

B.Pharma (Sem.-4)

MEDICINAL CHEMISTRY-I

Subject Code : BP-402T

M.Code : 75844

Date of Examination : 13-12-2025

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Explain the Terms :**

   a) Differentiate between classical and non-classical bioisostere.

   b) How is epinephrine synthesized from norepinephrine?

   c) Write chemical synthesis of Barbital.

   d) Give chemical structure and uses of Methyldopa.

   e) Draw chemical structures of Triclofos sodium and Zolpidem.

   f) Give the structure and specific medicinal use of Clozapine.

   g) What are dissociative anaesthetics? Give examples.

   h) What are the medicinal uses of Valproic acid and Acetaminophen?

   i) Give structure and medicinal uses of Diclofenac,

   j) Write chemical synthesis of Ethosuximide.

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SECTION - B

2. What are beta adrenergic blockers? Write SARof beta blockers. Write synthesis of propranolol.

3. Discuss ring analogues of Phenothiazine's and Fluorbutyrophenones as antipsychotics.

4. Classify sedative and hypnotics. Write in detail SAR benzodiazepines as sedative hypnotics.

SECTION - C

5. Give an account on SAR of morphine analogues.

6. Discuss biosynthesis and catabolism of acetylcholine

7. Write a brief note on cholinesterase inhibitors.

8. Classify anticonvulsants Discuss hydantoins as anticonvulsant drugs.

9. Discuss heteroarylacetic acid derivatives as anti-inflammatory agents.

10. Discuss in detail medicinal chemistry of ultrashort acting barbiturates.

11. Discuss in detail adrenergic receptors and their distribution.

12. Comment upon the role of chelation and partition coefficient on biological activity.

13. Comment on Phase I metabolic reactions in drug metabolism.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.